Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building full-stack healthcare and data-driven products using Python, TypeScript, SQL, React, REST APIs, testing, documentation, and secure workflow design.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **full-stack grading workflows** with React and Flask, handling 10,000+ daily requests while integrating uploads, OCR processing, persistence logic, and model-backed evaluations for instructor-facing product experiences.
- Debugged **runtime issues and release blockers** across upload, OCR, persistence, and grading features, reducing feedback turnaround by **60%** while improving issue isolation, testing confidence, and production readiness.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **backend services and data workflows** for 500+ users, translating stakeholder requests into APIs, validation logic, schema updates, and maintainable release pipelines across cloud-hosted systems and product-facing operations.
- Improved **system performance and production reliability** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** while strengthening debugging, traceability, and support.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **cross-platform APIs and ETL services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** while supporting data integrations, transformation workflows, and dependable release coordination.
- Maintained **20+ automated validation suites** with Jenkins, PyTest, and Mockito, increasing CI reliability by **68%** while strengthening defect isolation, repeatable testing, and cross-team delivery across production workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **REST APIs, WebSockets, and responsive interfaces** over SQL/NoSQL persistence, supporting 2M+ synchronized records across operational workflows while improving service coordination, reliability, and user-facing product responsiveness overall.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting speed, validation coverage, and engineering handoffs during iterative feature delivery.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask APIs and web workflows** using Firebase Cloud Messaging for 50K+ records, delivering responsive application behavior while supporting webhook integrations, notifications, and documented client-facing release processes.
- Integrated **payment and notification features** with maintainable backend logic and technical documentation, reducing manual handling by **70%** while improving usability, operational coordination, and support across releases consistently.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
Healthcare & Security: secure workflow design, consent-aware systems, cryptography familiarity, encryption concepts, sensitive-data handling
Full Stack: React, Next.js, Flask, REST APIs, WebSockets, HTML/CSS, responsive UI, technical documentation
Data & Delivery: PostgreSQL, MySQL, MongoDB, AWS Glue, S3, Git/GitHub, Docker, Jenkins, CI/CD, debugging, unit testing, integration testing

Projects

HMIS Live | *MERN, Twilio, Sentry, TypeScript, ABDM APIs*
- Built a **full-stack healthcare platform** with appointment workflows, consent-based access, notifications, and monitoring, reducing administrative workload by **30%** while improving internal coordination and operational visibility across teams.

Comprehensive Analysis of Blockchain for User Authentication | *Cryptography, Blockchain Authentication*
- Designed a **blockchain-authentication model** for secure user access, reducing repeated key-verification steps by **50%** while exploring tamper-resistant authentication flows and stronger protection for sensitive records in practice.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a **scalable fintech platform** supporting 50K+ users and 1M+ transactions with real-time synchronization, secure integrations, CI/CD-backed deployment workflows, and operational reliability across distributed application components overall.